EchoEmotion

Speech Emotion Recognition (ECL443 Course Project)

Deep Learning · Transfer Learning
Custom Dataset Evaluation

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Problem Statement

Human speech carries rich emotional cues that are critical for natural human-computer interaction. Current voice systems are largely emotion-agnostic.

This project develops an automated Speech Emotion Recognition (SER) system capable of classifying emotions from raw audio using deep learning.

Key challenges addressed:

- Speaker variability & accent diversity
- Cross-dataset domain shift (lab → real-world)
- Robust generalisation to unseen sentences & speakers
- Efficient adaptation without full retraining

Datasets & Baseline Approaches

RAVDESS

1,440 files

8 emotions · Phase 1 baseline

+ TESS

2,800 files

7 emotions · Phase 1 baseline

RAVDESS +TESS+CREMA-D

11,090 files

6 emotions · Approach 2

RAVDESS +TESS+CREMA-D+ SAVEE

12,000+ files

8 emotions · Approach 1 (final)

Feature Extraction (used in 1D CNN Approach)

ZCR	Zero-Crossing Rate – rhythm & noise
RMSE	Root Mean Square Energy – loudness
MFCC	40 Mel-Frequency Cepstral Coefficients
Aug ×3	Noise · Pitch Shift · Time Stretch

Baseline Model Comparison (RAVDESS /TESS, Phase 1)

Model	Test Acc	Notes
SVM (MFCC)	**~62%**	Classic ML baseline
Random Forest	**~58%**	Classic ML baseline
CNN (Mel-spectrogram)	**~65%**	DL – first upgrade
CNN + LSTM	**~70%**	Temporal context added
CNN + BiLSTM + Attention	**76.2%**	**Best RAVDESS model ✓**

Model Architectures

Approach 1 — CNN (Keras / TF)

Datasets: RAVDESS + TESS + CREMA-D + SAVEE | ~12,000 files | 8 emotions

Input — ZCR + RMSE + MFCC (2376-dim feature vector)

↓

Conv1D ×3 — Conv1D → BatchNorm → MaxPool → Dropout

↓

Dense ×2 — Dense(256) → Dense(128) + BatchNorm + Dropout

↓

Output — Dense(8, Softmax) — 8 emotion classes

Params: ~7.19M · Optimizer: Adam · Batch: 64

Approach 2 — CNN+BiLSTM+Attention (PyTorch)

Datasets: RAVDESS + TESS + CREMA-D | ~11,090 files | 6 emotions

Input — Mel-spectrogram (64 mel bins × 93 time steps)

↓

CNN ×3 — Conv2D 3×3 → BatchNorm → MaxPool → Dropout(20%)

↓

BiLSTM ×2 — BiLSTM(256) → BiLSTM(128) · Dropout(20%)

↓

Attention — Multi-head self-attention (4 heads) + BatchNorm

↓

Output — FC layers → Dense(6, Softmax) — 6 emotion classes

Params: ~2.5M · Optimizer: Adam + Cosine LR · Batch: 16

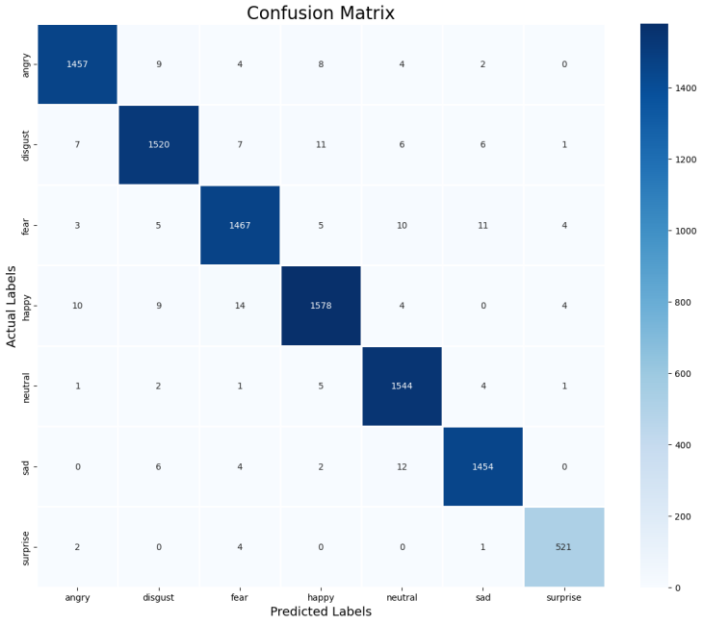
Training Results & Experiments

Approach 1 — Multi-Dataset CNN

98.05%
Best Test Acc

98%
Macro F1 (R7)

98.06%
Peak Val Acc

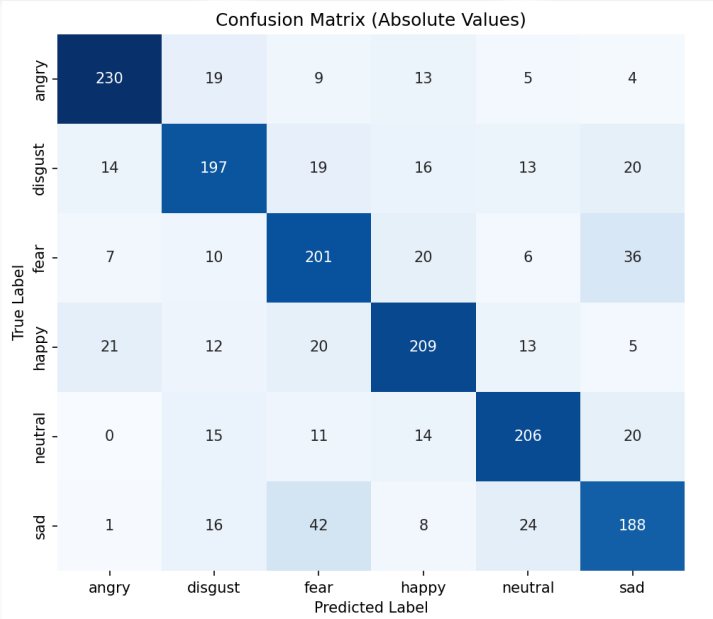


Approach 2 — CNN+BiLSTM+Attention

76.2%
RAVDESS Test Acc

75.8%
Macro F1

78.5%
Best Val Acc



Custom Dataset Testing & Domain Adaptation

Custom Test Dataset

- 80 WAV files · 5 Actors
- 4 Emotions: Neutral, Happy, Sad, Angry
- Seen vs Unseen sentences
- Natural speech, varied acoustics
- Different device & room condition

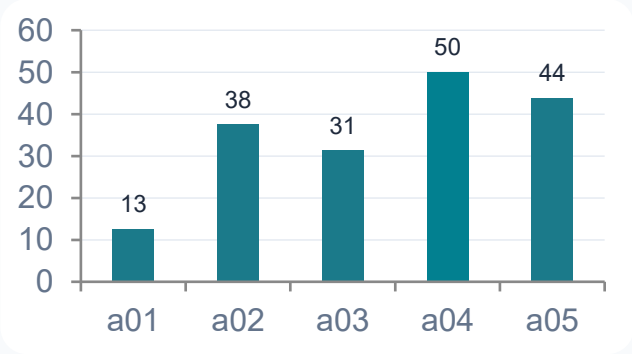
Why Accuracy Dropped — Domain Shift

Training data (RAVDESS/TESS/CREMA-D):
Studio-quality · North American accent · Exaggerated acted emotions
Custom data:
Natural speech · Varied accents · Different room acoustics · Unknown device
Emotion Mismatch:
· Sad and Neutral (and happy sometimes) of some samples of actor have similarity
· Uncontrolled audio samples, Human Accuracy is also less for some of these samples.

Accuracy Across Methods & Experiments

Method	Accuracy	Remark
Raw Inference (Approach 1 CNN)	22.5%	Severe domain mismatch
Label Mapping (7-class → 4-class)	30.0%	+7.5% gain
Probability Merging	28.75%	Marginal improvement
Frozen Head Fine-Tuning	27%	Limited without unfreezing
Approach 2 (CNN+BiLSTM) on Custom	35.0%	Best baseline on custom set
Partial Unfreeze Fine-Tuning ✓	**Best Practical**	**Recommended path forward**

Actor-wise Accuracy (Approach 2 Test)

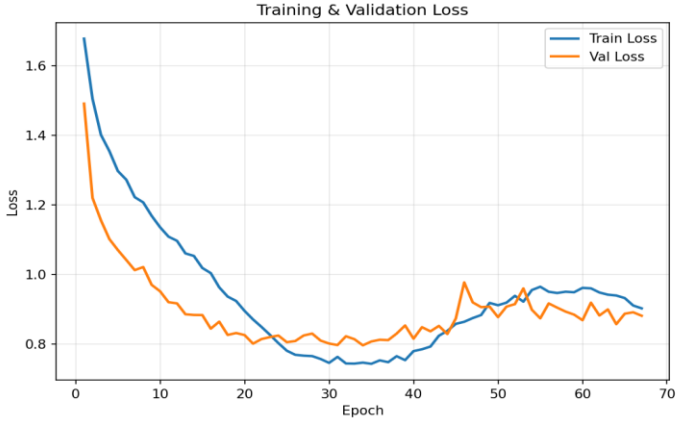


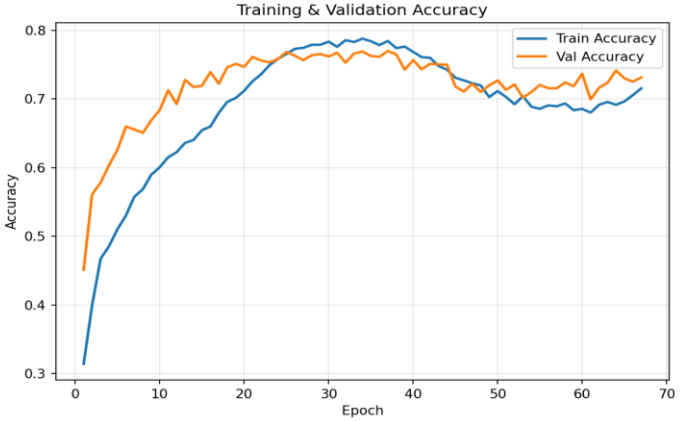
Seen sentences: 42.5% **Unseen: 27.5%**

Training Curves

Approach 1 — Multi-Dataset CNN

Approach 2 — CNN+BiLSTM+Attention





Key Findings & Recommendations

01 Benchmark ≠ Real-World

Models trained on studio datasets (RAVDESS, TESS) achieved 73–76% on standard splits but dropped to 22–35% on custom speakers, confirming the domain shift problem in SER.

02 Architecture Matters

CNN+BiLSTM+Attention (PyTorch, 2.5M params) outperformed plain CNN+Dense (7.19M params) both in validation accuracy and in custom-set generalisation, with fewer parameters.

03 Transfer Learning is Key

Label mapping gave +7.5% on custom data. Partial-unfreeze fine-tuning on custom speakers is the recommended path - best accuracy improvement without full retraining.

Future Work

- Expand custom dataset - more speakers, natural/diverse emotions
- Speaker-independent cross-validation for fair generalisation testing
- Explore Transformer / wav2vec2.0-based architectures
- Apply aggressive SpecAugment + waveform augmentation on custom data
- Automated hyperparameter search around the best run (R7) regime